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|----------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------|
| Program : <b>Diploma in Electronics Engineering / Electronics and Communication Engineering / Biomedical Engineering / Micro Electronics</b> |                                       |
| Course Code : <b>6201B</b>                                                                                                                   | Course Title: <b>Embedded Systems</b> |
| Semester : <b>5 / 6 / 5 / 6</b>                                                                                                              | Credits: <b>4</b>                     |
| Course Category: <b>Program Core / Elective / Elective</b>                                                                                   |                                       |
| Periods per week: <b>4 (L:3, T:1, P:0)</b>                                                                                                   | Periods per semester: <b>60</b>       |

### Course Objectives:

- Introduce the importance of embedded systems
- Introduce ATMEGA 32 microcontroller and its suitability for embedded systems
- Familiarize embedded C programming
- Familiarize serial communication and data transfer between embedded systems

### Course Prerequisites:

| Topic                                  | Course code | Course name                            | Semester |
|----------------------------------------|-------------|----------------------------------------|----------|
| Basics of programming concepts         | 3045        | Programming in C<br>Python programming | 3        |
|                                        | 4049        | Python programming                     | 4        |
| Basics of Microcontroller architecture | 4041        | Microcontroller and Application        | 4        |

### Course Outcomes:

On completion of the course, the student will be able to:

| CO <sub>n</sub> | Description                                                                    | Duration (Hours) | Cognitive Level |
|-----------------|--------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | Explain the basics of embedded systems and its architecture                    | 13               | Understanding   |
| CO2             | Make use of AVR Microcontrollers to develop embedded programs using embedded C | 16               | Applying        |
| CO3             | Make use of AVR microcontroller to interface with various peripheral devices.  | 19               | Applying        |

|     |                  |    |               |
|-----|------------------|----|---------------|
| CO4 | Familiarize RTOS | 10 | Understanding |
|     | Series Test      | 2  |               |

### CO - PO Mapping

| Course Outcomes | PO1 | PO2 | PO3 | PO4 | PO5 | PO6 | PO7 |
|-----------------|-----|-----|-----|-----|-----|-----|-----|
| CO1             | 2   |     |     |     |     |     |     |
| CO2             | 3   | 3   |     |     |     |     |     |
| CO3             | 3   | 3   |     |     |     |     |     |
| CO4             | 3   |     |     |     |     |     |     |

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

### Course Outline:

| Module Outcomes | Description                                                                                    | Duration (Hours) | Cognitive Level |
|-----------------|------------------------------------------------------------------------------------------------|------------------|-----------------|
| CO1             | <b>Explain the basics of embedded systems and its architecture</b>                             |                  |                 |
| M1.01           | Describe embedded system, illustrate difference from general purpose computer                  | 2                | Understanding.  |
| M1.02           | Classify embedded systems, explain application areas and summarize purpose of embedded systems | 2                | Understanding   |
| M1.03           | Distinguish Hardware and software components of embedded systems                               | 2                | Understanding   |
| M1.04           | Describe the basic blocks in a typical embedded system                                         | 2                | Understanding   |
| M1.05           | Describe Memory, Sensors, Actuators and I/O sub-systems                                        | 2                | Understanding   |
| M1.06           | Distinguish Communication Interfaces – On board and external interfaces                        | 3                | Understanding   |

#### Contents:

**Embedded Systems** - Definition, difference from general purpose computers - Classification of embedded systems, Application areas,

**Components of embedded system** hardware, and Software embedded into the system

**Architecture of embedded system** – Building blocks of an embedded system , Core of embedded system – categories, Memory –ROM and RAM, Sensors, actuators and I/O sub

systems (LED, Opto coupler, Relay, Stepper motor). On board (I2C, SPI, UART) and external communication interfaces (RS 232, USB, Bluetooth, Wifi)

|            |                                                                                                                                              |   |                |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------|---|----------------|
| <b>CO2</b> | <b>Make use of AVR Microcontrollers to develop embedded programs using embedded C</b>                                                        |   |                |
| M2.01      | Familiarize AVR controllers family members and criteria to select a microcontroller                                                          | 2 | Understanding. |
| M2.02      | Explain block diagram of Atmega32 and its blocks                                                                                             | 2 | Understanding  |
| M2.03      | Illustrate Registers, Memory organization, Status register, Program counter, I/O ports and its registers, Interrupts, priority of interrupts | 3 | Understanding. |
| M2.04      | Illustrate Timers in AVR                                                                                                                     | 3 | Understanding. |
| M2.05      | Develop embedded C programs for logic operations, data conversions and I/O operations.                                                       | 2 | Applying.      |
| M2.06      | Develop embedded C programs for time delay, time delay calculation. Timer programming and timer interrupts handling                          | 2 | Applying.      |
| M2.07      | Develop Programs to handle external hardware interrupts and programming based on priority of interrupts                                      | 2 | Applying.      |
|            | Series Test - I                                                                                                                              | 1 |                |

**Contents:**

**AVR Microcontroller Architecture** - Comparison of AVR family members and Selection of a microcontroller, ATmega32- Simplified Block diagram of ATmega32 microcontroller - Registers, - data memory - I/O memory -SFRs - internal data SRAM, Status register, Program Counter and Program ROM space, I/O ports, Registers associated with I/O ports, Timers-0,1,2 (Block level) associated registers, Interrupts.

**AVR programming using embedded C:** Data types, I/O programming, logic operations - data conversion programs - time delays- programming of timers 0 - timer 1 - timer2, AVR interrupts - programming of timer interrupts - programming external hardware interrupts - interrupt priority in AVR microcontroller.

|            |                                                                                                 |   |               |
|------------|-------------------------------------------------------------------------------------------------|---|---------------|
| <b>CO3</b> | <b>Make use of AVR microcontroller to interface various peripheral devices using embedded C</b> |   |               |
| M3.01      | Illustrate the need for interfacing, and types of interfacing devices                           | 2 | Understanding |
| M3.02      | Illustrate the interfacing of LED, Push button, Relay and Optocoupler with AVR                  | 2 | Applying      |
| M3.03      | Illustrate Sensors and Seven segment Display interfacing with AVR                               | 2 | Applying      |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                       |   |               |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|---|---------------|
| M3.04                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Make use of AVR to realize LCD and Keyboard interfacing                               | 3 | Applying      |
| M3.05                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Make use of AVR microcontroller to interface DC motor, Servo motor and stepper motor. | 3 | Applying      |
| M3.06                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Make use of AVR microcontroller to interface RTC and ADC.                             | 2 | Applying      |
| M3.07                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Realize I2C interfacing with AVR                                                      | 3 | Applying      |
| M3.08                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Understand SPI interface with AVR                                                     | 2 | Understanding |
| <b>Contents:</b><br><b>Interfacing</b> of LED, Push button, Relay, Opto Coupler, Sensors (Temperature sensor, IR sensor) Seven segment Display, LCD, Keyboard Interfacing, Motor (DC, Servo, Stepper), RTC Interfacing, ADC interfacing<br>On-board communication interfaces with AVR – I2C interfacing(like real time clock interfacing)and basics of SPI interfacing with AVR.                                                                                  |                                                                                       |   |               |
| <b>CO4</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <b>Familiarize RTOS</b>                                                               |   |               |
| M4.01                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Describe Kernel,Operating System Architecture and types                               | 2 | Understanding |
| M4.02                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ExplainRTOS Kernel functions                                                          | 3 | Understanding |
| M4.03                                                                                                                                                                                                                                                                                                                                                                                                                                                             | ListSelection criteria for RTOS                                                       | 2 | Remembering   |
| M4.04                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Outline Micro C/OS-II and its services                                                | 2 | Understanding |
| M4.05                                                                                                                                                                                                                                                                                                                                                                                                                                                             | List popular Real Time Operating Systems.                                             | 1 | Remembering   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Series Test - II                                                                      | 1 |               |
| <b>Contents:</b><br><b>Operating system basics:-</b> Kernel, types of operating systems- GPOS, RTOS. <b>Real time operating systems:-</b> Tasks, process, threads, multiprocessing and multi-tasking, task scheduling, types, threads and process scheduling, task communication, task synchronization, device drivers.<br>List Selection criteria for RTOS.<br>Overview of Micro C/OS-II and its services.<br>List popular Real Time Operating Systems (any 10). |                                                                                       |   |               |

**Text / Reference:**

| <b>T/R</b> | <b>Book Title/Author</b>                                                                                                             |
|------------|--------------------------------------------------------------------------------------------------------------------------------------|
| T1         | K.V. Shibu, Introduction to Embedded Systems, 2e, McGraw Hill Education India, 2016.                                                 |
| T2         | Rajkamal, Embedded Systems Architecture, Programming and Design, TMH, 2003                                                           |
| R1         | Muhammad Ali Mazidi, SarmadNaimi, &SepehrNaimi, The AVR Microcontroller and Embedded Systems Using Assembly and C, Pearson Education |
| R2         | Michael J. Pont, Embedded C, Pearson Education, Second Edition                                                                       |

**Online Resources:**

| <b>Sl.No</b> | <b>Website Link</b>                                                                                                                                                                                                                     |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1            | <a href="https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-beginners/">https://www.studyelectronics.in/embedded-programming-tutorial-chapter-1-beginners/</a>                                                     |
| 2            | <a href="https://www.tutorialspoint.com/embedded_systems/index.htm">https://www.tutorialspoint.com/embedded_systems/index.htm</a>                                                                                                       |
| 3            | <a href="https://embeddedschool.in/avr-microcontroller-programming/">https://embeddedschool.in/avr-microcontroller-programming/</a>                                                                                                     |
| 4            | <a href="https://learn.sparkfun.com/tutorials/i2c/all">https://learn.sparkfun.com/tutorials/i2c/all</a>                                                                                                                                 |
| 5            | <a href="https://www.best-microcontroller-projects.com/i2c-tutorial.html">https://www.best-microcontroller-projects.com/i2c-tutorial.html</a>                                                                                           |
| 6            | <a href="https://www.electronicwings.com/avr-atmega/atmega1632-i2c">https://www.electronicwings.com/avr-atmega/atmega1632-i2c</a>                                                                                                       |
| 7            | <a href="https://www.electronicwings.com/avr-atmega/atmega1632-spi">https://www.electronicwings.com/avr-atmega/atmega1632-spi</a>                                                                                                       |
| 8            | <a href="https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.html">https://www.intel.com/content/www/us/en/docs/programmable/683525/21-3/overview-of-the-microc-os-ii-rtos.html</a> |